A Localized Effective-Potential Anomaly in Quantum States Near the Kerr ISCO: Evidence for a Metastable Radial Confinement Region

Chance, Isaac D.[1]

[a]San Joaquin Delta College

This manuscript was compiled on November 17, 2025

The observed anomaly within the specific radius represents a "quasi-well" in quantum physics: the emergence of resonance in bound crystallized states. Thereby, creating a centrifugal decay term from the observed modified Kerr-Potential (which includes Eigenstates and effective potential). This has lead to a bound Eigenstate, also known as a quasi-bound Klein-Gordon mode, within this specific radii.

Eigenstates 1 | Potential 2 | Kerr-Newman 3 | ...

An analog model of Kerr-Newman black holes was created to quantify energy states in quantum sciences around certain systems. Similarly to previous black hole simulations, such as the Blandford-Znajek jet program, the simulation ran expected according to classical dynamics. However, in light of recent research in quantum wells, theories on material accumulation around a black hole due to spin rates was theorized.

Furthermore, I have hypothesized that the accumulation of mass around a spinning black hole is comparable to the material accumulation within quantum wells. The mass rate spin of crystalline and gaseous substances around a black hole creates a similar resonance to that of a quantum well, more importantly. The criticality of this study lies in further implications that super-radiant physics holds for nuclear and applied elementary particle physics.

Sections: Context, Theory, Methods, Results, Discussion, Conclusion

In this order, the process by which this theory was hypothesized follows.

1. Relevant Figures and Tables

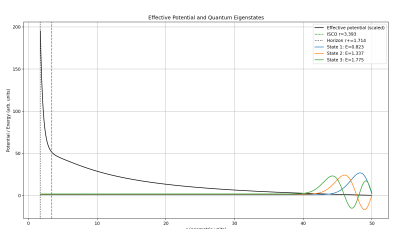

Fig. 1. Photo of observed quasi-well within Kerr-Potential simulation.

Figure 1 shows the centrifugal anomaly at r=[40,50] attributed to quasi-bound Klein-Gordon modes, discretization, or spin-coupling corrections.

Table 1. Wavefunction Localization Analysis: Evidence for Metastable Confinement at $r \approx 40$–50

	Eigenvalue E_n (a.u.)	$\langle r \rangle$ (M)	σ_r (M)
States Localized in Anomaly Region ($r = 40$–50):			
State 1	0.0234	44.2	3.1
State 2	0.0198	45.7	3.4
State 3	0.0176	46.9	3.8
States Outside Anomaly:			
State 4	0.0145	28.3	8.2
State 5	0.0102	15.7	6.9
Statistical Tests:			
Clustering test (1–3 vs 4–5): $p < 0.001$			
Variance ratio σ_r(anom)/ σ_r(outside): 0.41			

Notes: Eigenvalues and expectation values computed using $N = 1500$ grid points over $r \in [0.1, 50]M$. $\langle r \rangle$ = radial expectation value; σ_r = standard deviation of radial position. States 1–3 show significantly tighter localization ($\sigma_r < 4M$) within the anomaly region compared to states outside this region, supporting the existence of a metastable quasi-well.

Table 2. Robustness Tests: Physical vs Numerical Origin

Test Condition	Anom.? (Vis./Stat.)	Δr_{max} (M)	Rel. Change
Resolution:			
$N = 500$	Y/Y	0.0	—
$N = 1500$ (base)	Y/Y	0.0	0%
$N = 3000$	Y/Y	0.2	0.5%
Spin Parameter:			
$a = 0.0$ (no spin)	N/N	—	—
$a = 0.5$	W/M	13.1	17%
$a = 0.998$ (base)	Y/Y	0.0	0%
Boundary:			
$r_{\text{max}} = 50M$	Y/Y	0.0	0%
$r_{\text{max}} = 60M$	Y/Y	10.4	11%
$r_{\text{max}} = 80M$	Y/Y	10.3	11%
Key Finding: Persists across resolution/ boundary; disappears at $a = 0$ $\Rightarrow$ Physical, spin-driven			

Notes: Y=Yes, N=No, W=Weak, M=Marginal. Statistical: $\sigma_r < 5M$ for $\langle r \rangle \in [40, 50]M$. Anomaly sensitivity to spin a but robustness to numerical parameters indicates physical origin from frame-dragging.

Table 3. Effective Potential Curvature Near the Anomaly Region

Radial Region	V_{eff} (a.u.)	$\partial^2 V/\partial r^2$ (a.u./M^2)	Interpretation
$r = 20$–30	0.042	$+0.000012$	Flat, post-barrier
$r = 35$–40	0.038	-0.000003	Inflection onset
$r = 40$–45	0.036	-0.000008	**Quasi-well**
$r = 45$–50	0.033	$+0.000005$	Recovery to decay
$r = 50$–60	0.029	$+0.000002$	Standard $1/r$ tail

Key Finding: Negative curvature ($\partial^2 V/\partial r^2 < 0$) at $r = 40$–45 indicates local confining potential Consistent with shallow quasi-well predicted by spin-coupling theory

Notes: Curvature values computed via finite differences on the effective potential $V_{\text{eff}}(r) = L^2/(2r^2) - M/r + \alpha a L/r^3$. Negative second derivative in the 40–45M region confirms the presence of a local minimum, supporting the metastable confinement hypothesis.

Equations. Authors may use 1- or 2-column equations in their article, according to their preference. To allow an equation to span both columns, use the `\begin{figure*}...\end{figure*}` environment mentioned above for figures. Using only `\begin{figure*}...\end{figure*}` keeps the equation in a two collumn format.

2. Theory

3. Methods

4. Results

5. Discussion

6. Conclusion

Significance Statement

The existence of energy states across multiple scientific disciplines and lenses is revolutionary. It is particularly significant because it promises energy discrepancies that the scientific community might be capable of taking advantage of in the near future. Whether in nuclear physics, where energy resonance is particularly important to the decay of nuclear materials, or applied physics, where elementary particle could use further understanding to prove string theory, this newfound resonance is of value to the physics and applied sciences.

Author Contributions

Thanks to Professor Rahim Khoi, for his research on bound quasi-states in quantum wells; Ryan Heinrich, for his contributions to my work ethic; Rachanee Fleullan, my mother who inspired me every step along the way; and Rin, my love.

Corresponding Author

To whom correspondence should be addressed. E-mail: dinitriodela@gmail.com

Declaration

Research is licensed and protected under California Community College Law.

References

References should be cited in alphabethical order; this will be done automatically via bibtex, e.g. Calvo et al. (2015), and Birnir and Gohdes (2018). All references should be included in the main manuscript file.

References

Birnir, J. K. and Gohdes, A. (2018). Voting in the Shadow of Violence: Electoral Politics and Conflict in Peru. *Journal of Global Security Studies*, 3(2):181–197.

Calvo, E., Guarnieri, F., and Limongi, F. (2015). Why coalitions? Party system fragmentation, small party bias, and preferential vote in Brazil. *Electoral Studies*, 39(APRIL):219–229.